3.2 Implicit Differentiation

Name _____ Period ____ Date _____

AP Calculus 3.2 (30-minute plan) · Student sheet

I can…

- find dy/dx by implicit differentiation.
- find the slope and the tangent line of an implicit curve at a point.
- use the Power Rule with rational exponents.

Vocabulary · fill in the missing word

Implicit function — a function y = f(x) defined by an _____ in x and y that is not solved for y

Implicit differentiation — differentiate both sides with respect to x, _____ y as a function of x, then solve for $\frac{dy}{dx}$

Power Rule, rational exponents — $\frac{d}{dx}\,x^{p/q}$ = _____

Part 1 · Notes and examples (fill in each note, then work the example)

Implicit differentiation: differentiate both sides with respect to x. Each time you differentiate a y-term, multiply by _____.

Example 1

Find $\frac{dy}{dx}$ for $x^2 + y^2 = 25$, and the slope at $(3, -4)$.

Solve for dy/dx: collect the dy/dx terms on one side and _____.

Example 2

Find $\frac{dy}{dx}$ for $x^2 + xy + y^2 = 7$, and the slope at $(1, 2)$.

Second derivative: differentiate dy/dx again, then replace dy/dx with _____.

Rational exponents: d/dx x^(p/q) = _____.

Example 3

Find each derivative.

a. $y = x^{2/3}$

b. $y = \sqrt[3]{(x^2 + 1)}$

Part 2 · Partner practice

Partner practice

(AP® Practice) If $x^2 + y^3 = 9$, then $\frac{dy}{dx}$ at the point $(1, 2)$ is

(A) $-\frac{1}{6}$ (B) $-\frac{1}{3}$ (C) $\frac{1}{6}$ (D) $-\frac{2}{3}$